

ECEC state demand inputs: method

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1 Purpose

The simulator reads one file for each geography. The file holds six rows, one for each care type, and each row holds the annual hours of child care. This memo gives the source of those numbers.

The build covers five geographies: Kentucky (KY), New Jersey (NJ), New York (NY), New York City (NYC) and Virginia (VA). Section 4 lists every source by the agency that publishes it, with the address of each one. Source names in the text link to those addresses.

Why the build exists

The simulator must know the mix of child care in each state, and the survey that measures that mix cannot supply it. NSECE does not record the state of a family. It reports the Census region and nothing below it, so no filter recovers a state from it.

I therefore build the mix from the administrative records of each state, and I use NSECE only for the care that no administrative record shows. Two rules follow from that, and they hold across every step below.

- The build works at the state level. It reads county data one time, in step 1, and it adds the counties together immediately. No later step uses a county, so the build needs no county crosswalk and no county name matching.
- The build prefers a state measurement to a national one. It uses a national figure only where the state publishes nothing, and it says so each time it does.

The position of NYC

NYC is a geography of its own here, and every step below gives its figures. It is also inside NY, and the NY figures include it. The city appears twice because the New York City Department of Health and Mental Hygiene regulates its centers rather than the state agency that regulates everywhere else, and because the city is asked about on its own.

The two builds do not agree, and that is deliberate. Each one takes its own control total in step 2, and step 5 calibrates each one separately, so the five boroughs come out differently depending on which build produced them. Read one or the other. Do not add them together.

2 Results

The build counts two quantities and calculates a third as a remainder. Every row in the output divides one of the three.

	Center-based	Not center-based	Total	Share center
KY	81,326	80,604	161,930	50%
NJ	173,225	136,721	309,946	56%
NY	270,982	378,847	649,829	42%
NYC	132,065	169,911	301,976	44%
VA	134,434	128,566	263,000	51%

Care that is not center-based splits into a part the build counts and a part it calculates as a remainder.

	Licensed home care	Informal care
KY	4,756	75,848
NJ	1,987	134,734
NY	96,366	282,481
NYC	38,642	131,269
VA	5,377	123,189

NY and NYC hold far more licensed home care than the other three geographies, and the difference is real rather than an error. The New York Office of Children and Family Services registers a home provider that keeps three children or more, so care that leaves no trace elsewhere sits inside its register. Kentucky, New Jersey and Virginia register only larger providers. That care exists in those states too, but it falls into the remainder instead.

The figures below are counts of children. They are not record counts, and they are not weighted survey figures.

	KY	NJ	NY	NYC	VA
Center-Based	37,550	116,402	168,235	86,337	86,558
Unpaid Center-Based	43,776	56,823	102,747	45,728	47,876
Paid Home-Based	4,756	1,987	96,366	38,642	5,377
Other Unpaid	45,830	81,411	170,686	79,318	74,436
Other Paid	27,363	48,607	101,909	47,357	44,442
Unpaid Home-Based	2,655	4,715	9,886	4,594	4,311
Total	161,930	309,946	649,829	301,976	263,000

The output file holds annual hours, shown here in millions. Step 6.3 gives the hours for each child, which come from NSECE.

	KY	NJ	NY	NYC	VA
Center-Based	63.3	196.2	283.6	145.5	145.9
Unpaid Center-Based	62.9	81.6	147.6	65.7	68.8
Paid Home-Based	9.1	3.8	185.0	74.2	10.3
Other Unpaid	88.4	157.1	329.3	153.0	143.6
Other Paid	42.5	75.5	158.4	73.6	69.1
Unpaid Home-Based	5.6	10.0	21.0	9.7	9.1

3 The build

Step 1. Count the children

I read the Census population estimates for 2019, keep children of age 0 to 4, and add the counties together for each geography.

	Counties	Children 0–4, 2019
KY	120	272,610
NJ	21	514,690
NY	62	1,127,001
NYC	5	523,718
VA	133	505,477

NYC uses five counties from the same file: Bronx 100,199; Kings 182,926; New York 76,144; Queens 137,095; Richmond 27,354. The city holds 46 percent of the children in NY.

The build stops at age 4 for three reasons.

- The simulator stops there. Its ACS processing keeps children below age 5 in seven places, so a wider demand file would anchor the model against households it never sees.
- A universe of ages 0 to 5 raises each control total by 15 to 23 percent. That rise combines a population about 19 percent larger with a survey rate that moves, because five-year-olds use non-parental care less than younger children do. Most of them are in kindergarten, which the survey does not count as child care.
- Census publishes no county estimate for age 5 on its own. The build can count a state at that age, but it could only estimate NYC, so the city would stop being counted and start being modelled.

Step 2. Find the children in non-parental care

I read the NSCH household survey for 2019, keep the children of age 0 to 4, weight each child by the survey child weight, and take the share in any non-parental care. The product of that share and the population from step 1 is the control total, and every later count is a share of it.

	Children 0–4	Rate	In non-parental care	Records
KY	272,610	59.4%	161,930	147
NJ	514,690	60.2%	309,946	101
NY	1,127,001	57.7%	649,829	112
NYC	523,718	57.7%	301,976	—
VA	505,477	52.0%	263,000	121

- The control total is always a 2019 figure, because both of its inputs are 2019 data. This is what allows the build to read a register published in 2026 without ageing it backward: the register decides how care divides across types, and it never decides how much care there is.
- NYC has no records of its own, which is why its row shows a dash. It takes the rate of NY, because NSCH publishes no geography below the state. Every difference between the city and the state therefore has to come from the registers in step 3, and it does.

- This rate is the weakest number in the build. Each geography rests on about 120 families, which is a standard error of five to six percentage points on a figure that multiplies every later count. Section 4.1 explains why the build still uses a single survey year.

Step 3. Convert the registers into children

Every state licenses child care providers and publishes a register, and every register reports capacity rather than children. Three sub-steps convert one into the other.

3.1 Remove the school-age capacity

Each register mixes school-age capacity into its figures, and each one hides it differently.

- Kentucky’s Division of Regulated Child Care publishes no bulk file, so the build reads Child Care Aware of America’s republication of its licensed and certified registers. Neither register says which providers are centers and which are homes, so the build applies the state licensing rule: a licensed home keeps 12 children or fewer. That gives 1,609 centers and 335 homes. Providers serving school-age children only are dropped. A mixed age band scales by a ratio between 0.487 and 0.827, and those ratios are measured inside the register itself, by comparing the mean capacity of a band against the same band without school-age children.
- The New Jersey Department of Children and Families (DCF) publishes its register of licensed centers, with an age range for each one. The build drops the 1,035 centers serving ages 6 to 13 only, and scales a mixed range by 0.820. That scale is also measured inside the register, from two band comparisons that give 0.8224 and 0.8183.
- The New York Office of Children and Family Services (OCFS) publishes its register with capacity broken out by age, which is the clean case. Infant plus toddler plus preschool is the under-five capacity exactly, so no scale is needed. Homes report a single total, but the register publishes their school-age capacity separately and the build subtracts it.
- NYC draws on both agencies. Its centers come from the New York City Department of Health and Mental Hygiene (DOHMH), whose permit file gives one capacity figure per permit and covers children below age 6 by construction. Its homes come from the OCFS register and take the same subtraction as NY.
- The Virginia Department of Social Services (VDSS) writes the age range as text in its register, so the build parses it into months, drops any provider starting at age 5 or later, and scales a center reaching past age 6 by 0.60. VDSS appends a contact block to that field, and the build strips it before parsing: the appended phone number carries a hyphen, and an unstripped string makes the upper bound unreadable. Section 4.2 gives what that cost when it was missed. With the field read correctly, a span ending below age six holds 0.527 of one running to nearly thirteen in Child Day Centers and 0.604 in Religious Exempt centers, pooling to 0.540, so 0.60 is the conservative end of the measured range. Homes take no scale at all, because VA caps a family day home at twelve children whatever ages it serves.

3.2 Move the register back to 2019

Three of the registers are 2026 exports and have to be brought back to the base year.

- NY scales each provider type separately, against the OCFS series of provider counts and capacity by modality covering 2015 to 2025. Group family day care scales to 0.86 of its 2026 size, and OCFS centers to 0.93. The DOHMH centers that NY also carries scale at 1.00, as they do in the NYC build. Homes have closed in numbers that centers have not, so one blanket factor would distort the mix rather than correct it.
- NYC takes its own factors from the same OCFS series, which publishes the city separately from the rest of the state. Group family day care scales to 0.84 in the city, and city centers scale to 1.00, since the city register held almost the same capacity in 2019 as it does now.
- VA scales by one statewide factor of 0.90. The build computes it from VDSS’s Child Care Data Matrix for June 2020, as that report’s child care slots divided by the current VDSS register’s capacity.
- KY and NJ need no correction, because their registers already sit at the base year. The Child Care Aware republication is a September 2018 snapshot, and the DCF register is frozen at May 2019.

3.3 Convert capacity into children

Capacity is a legal maximum, so real enrollment is lower and something has to bring it down.

- KY needs no fill rate, because it measured its own enrollment. The 2019 Kentucky Workforce Study prints capacity and enrollment side by side in Table 20, giving 42 children for each center and 9 for each home. This is the preferred method, and it is why 1,609 centers yield only 67,578 children.
- NJ measured its own fill rate. The 2021 NJ Child Care Market Rate Survey puts its centers at 67 percent full in its 2020 column.
- NY, NYC and VA have published no fill rate, so they borrow the national figures: 90 percent for a center, from the ACF report on center-based providers, and 60 percent for a home, from the ACF report on home-based providers.
- NY and NYC homes use 80 percent instead of 60. The 60 percent figure is itself a product of a fill rate and an under-five share, and the OCFS register already removed the school-age capacity in sub-step 3.1. Applying the full figure would remove school-age children a second time, so those homes take the fill rate alone.

		Providers	Under-5 capacity	Children
KY	Centers	1,609	107,369	67,578
KY	Homes	335	2,115	3,015
NJ	Centers	3,128	251,159	169,106
NY	Centers, state	2,073	153,053	138,191
NY	Centers, city	2,310	141,300	127,580
NY	Homes	11,918	117,401	93,381
NYC	Centers	2,310	141,300	127,580
NYC	Homes	4,580	46,057	36,634
VA	Centers	3,186	188,600	170,287
VA	Homes	1,935	12,633	7,604

Providers and capacity come from the five registers named in sub-step 3.1 – Child Care Aware for KY, DCF for NJ, OCFS for NY, DOHMH for NYC centers, VDSS for VA. Children come from the conversion in sub-step 3.3.

NY appears twice in that table because two agencies regulate child care there. OCFS licenses providers across all 62 counties, but it licenses no center inside New York City, because DOHMH permits those itself under Article 47 of the city Health Code. The OCFS register therefore shows zero centers in the five boroughs, and that is the correct result rather than a failed match. OCFS does license home providers inside the city, so NYC reads centers from DOHMH and homes from OCFS.

DCF licenses centers only, so NJ has no register of home providers at all. Its home care comes from the same market rate survey as its fill rate, which counted 1,396 children below age 5 across 913 responding providers. The build raises that to the 1,254 registered providers in the state, giving 1,917 children. The 1,987 in the Results tables is that figure after step 5 scales it. Every count in Results is post-calibration; every count in the table above is pre-calibration.

Step 4. Add the programs that count their children

Head Start, Early Head Start and public pre-K publish actual enrollment, so the build takes the published count rather than estimating one.

	Pre-K	Head Start	Early Head Start	Total
KY	28,465	12,307	3,004	43,776
NJ	41,305	11,940	3,578	56,823
NY	54,451	36,863	11,433	102,747
NYC	26,717	13,073	5,938	45,728
VA	33,790	11,579	2,507	47,876

- Pre-K comes from the school-level records of the NCES Common Core of Data for 2018–19, counted school by school.
- Head Start comes from the NIEER State of Preschool 2019, and Early Head Start from the NIEER report on Head Start. The two NIEER publications disagree slightly on Head Start: the yearbook gives KY 12,307 and the Head Start report 12,163. The build uses the yearbook figure for consistency with the other four geographies, so it sits 144 children, or 1.2 percent, above the pair that reconciles exactly to the published federal total of 15,167.
- NYC takes its Head Start figure as a share of NY, using the ACF service location file: 7,194 funded places sit in the five boroughs against 20,285 in the state, so the city takes 35 percent of the state level.
- The build subtracts these children from the licensed center count rather than adding them. A Head Start center usually holds a state licence, so step 3 already counted its children, and adding them would count them twice.
- A published count does not move in step 5. The build treats it as a census, so the calibration falls entirely on the capacity-derived counts.
- This row is an upper limit everywhere. Head Start and state pre-K overlap in an unknown number of children, and no source publishes the size of that overlap.

Step 5. Calibrate the total to the survey

A register describes the relative distribution across care types well and the total badly: it puts far more children in formal care than any household survey finds. The build corrects the total with one factor and leaves the distribution alone.

1. Take the share of non-parental care that is formal. NSECE gives it by Census region: 56.5 percent in the Northeast and 53.2 percent in the South.
2. Multiply that share by the control total from step 2 to get the target.
3. Hold the program counts from step 4 fixed.
4. Scale every other count by one factor until formal care reaches the target.

	Scale factor	Formal	Informal	Informal share
KY	1.58	86,082	75,848	47%
NJ	1.04	175,212	134,734	44%
NY	1.03	367,348	282,481	44%
NYC	1.05	170,707	131,269	44%
VA	0.71	139,811	123,189	47%

The build computes each factor from the two totals, so nothing here is chosen by hand. This table is the primary check on everything above it, and it reads as follows.

- NJ, NY and NYC barely move. Their counts landed within two percentage points of the survey before the build touched them, which means three registers kept by three different agencies – DCF, OCFS and DOHMH – agree with one independent national survey. That agreement is the strongest evidence here that the conversion in step 3 is sound.
- VA falls by 29 percent. The VDSS register is the most complete of the four, listing unlicensed and voluntarily registered family homes that no other state here records, and it also borrows the national fill rate. Both push its count of formal care up, so the correction pushes it back down.
- KY rises by 58 percent, and this is the informative case. KY is the only state that measured how full its own centers are, and the answer was 57 percent against the 90 percent that NY and VA borrow. The borrowed figure is too high by more than half. KY moving up while VA moves down is exactly what that difference predicts, so the calibration is correcting an error that has been measured rather than absorbing one that has not.

Step 6. Complete the rows

6.1 Use no price split

An earlier version of this build divided center care into low-priced and high-priced. Nothing in the state data supports that division. A state publishes one price, so every state fell wholly on one side of the national boundary, and a care type left with no children then reads as unmeasured rather than as genuinely absent.

Center care is now a single row. It carries 1,686 hours for each child each year, which is the child-weighted mean across the two price bands that NSECE distinguishes nationally.

6.2 Calculate the care that no register records

Three of the six types appear in no register anywhere: unpaid relatives, nannies and informal arrangements. Unpaid relative care is not a business, files no return, and leaves no administrative trace of any kind.

The build subtracts all counted care from the control total and treats the remainder as this care, then divides that remainder three ways on the national pattern from NSECE. The remainder runs 44 to 47 percent of non-parental care across the five geographies, against 47 percent nationally in the same survey.

6.3 Convert children into hours

Each care type carries a measured hours figure from NSECE: the child-weighted mean weekly hours reported for the child's primary arrangement, multiplied by 50 weeks. The figures run from about 1,440 hours a year for program center care to about 2,120 for unpaid home care, so the hours column is not proportional to the children column.

4 Provenance

4.1 Why the rate uses one survey year

Pooling survey years would cut the sampling error on the non-parental rate by 40 to 47 percent, which would be the cheapest accuracy gain available. The build does not pool, because every input has to sit on the simulator's 2019 base year.

A pool of 2018, 2019 and 2020 centres exactly on 2019, so it is the one pool that passes the vintage test. It fails on content instead. NSCH collected its 2020 data through the pandemic, and the use of child care collapsed.

	2018	2019	2020
KY	57.3%	59.4%	45.8%
NJ	58.3%	60.2%	42.6%
NY	64.7%	57.7%	62.1%
VA	61.7%	52.0%	46.7%

All four rows come from NSCH. The fall in 2020 is a real change rather than sampling noise, and pooling it in would pull every total down by several points. A pool of 2018 and 2019 alone avoids that, but it centres on the middle of 2018 while every other input is 2019.

So the rate keeps its full sampling error, and the error is reported instead of being traded for a bias that nothing downstream could recover. The repository holds all three survey years, so anyone can repeat the test.

4.2 A parse defect in the Virginia register, and what it cost

VDSS writes the age range and a contact block into one field:

```
1 month - 12 years 11 months VDSS Contact:  
Tara K Martin: (804) 588-2312
```

The build reads the upper bound as the text after the last hyphen. In a string like that one, the last hyphen sits inside the telephone number, so the upper bound became 2312, which carries neither “year” nor “month” and parsed to zero months.

A provider whose upper bound reads zero looks like a provider that serves no school-age children, so the build kept 100 percent of its capacity. The rows affected mostly run from one month to twelve years and eleven months, which is the most mixed span the register holds.

	Affected
Rows	1,055
Licensed places	78,486
Share of the VDSS register	21.2%
Religious Exempt Child Day Centers	861
Voluntarily registered family day homes	173
Certified pre-schools	21

Stripping the contact block before parsing resolves every one of them, and it changes Virginia alone:

	Before	After
Mixed-span centers scaled	2,965	3,591
Center capacity below age 5	207,835	188,600
Center children	187,654	170,287
Calibration factor	0.62	0.71

Two things follow from this, and both matter more than the numbers.

The defect was invisible because the calibration hid it. Step 5 forces Virginia’s formal care to the regional share whatever the register says, so the extra 19,235 places were absorbed into a more extreme scale factor rather than into an obviously wrong total. What looked like Virginia having an unusually complete register was partly a parse failure. The register is still the most complete of the four, but less of the gap is real than the earlier figure implied.

The scale itself survived the fix. The 0.60 share was first measured on the uncleaned strings and happened to land in the right place. Measured on the cleaned field it is 0.527 for Child Day Centers and 0.604 for Religious Exempt centers, pooling to 0.540, so 0.60 remains defensible as the conservative end. The value did not change. What changed is that it now rests on a field the build reads correctly.

The build now stops if any Virginia age range parses to an upper bound of zero months, so the same failure cannot return silently if VDSS changes the field again.

4.3 Why the borrowed fill rates are too high

Two states have published their own figures, and both point the same way.

KY measured its centers at 57 percent full in 2019, against the 90 percent that NY, NYC and VA borrow. VA measured a related quantity in 2022, asking 1,968 providers what they can actually serve and comparing the answer to their licence: centers reported 73 percent of their authorized capacity and homes 91 percent, with 121,260 licensed places in that sample against 89,374 children the providers said they could take.

The build uses neither figure as a rate. KY’s applies to KY. VA’s is from 2022 and measures capacity rather than enrollment, so it would duplicate a correction the fill rate already makes. What

both establish is the direction and rough size of the error: licensed capacity overstates real supply by a quarter to a third, and the national fill rate does not remove enough of it.

4.4 Sources, by publishing agency

US Census Bureau

Population Estimates, county characteristics, vintage 2019, for children 0–4 by county.

<https://www2.census.gov/programs-surveys/popest/datasets/2010-2019/counties/asrh/cc-est2019-alldata.csv>

National Survey of Children’s Health, topical files for 2018, 2019 and 2020, for the non-parental care rate.

<https://www.census.gov/programs-surveys/nsch/data/datasets.html>

Administration for Children and Families

Part of the US Department of Health and Human Services. Center-based providers in 2012 and 2019, for the 90 percent center fill rate.

<https://acf.gov/opre/report/center-based-early-care-and-education-providers-2012-and-2019-counts-and>

Home-based providers in 2012 and 2019, for the 60 and 80 percent home figures.

<https://acf.gov/opre/report/home-based-early-care-and-education-providers-2012-and-2019-counts-and-characteristics>

Head Start service locations and funded places, for the NYC share.

https://s3foa.s3.us-east-1.amazonaws.com/HS_Service_Locations.csv

Head Start Program Facts for fiscal year 2019, used to check the KY program figures.

<https://acf.gov/ohs/data/program-facts>

ICPSR, University of Michigan

National Survey of Early Care and Education 2019, public-use files, study 37941. Supplies the formal share by region, the informal split, the hours for each care type, and the price bands behind the 1,686-hour figure.

<https://www.icpsr.umich.edu/web/ICPSR/studies/37941>

National Center for Education Statistics

Common Core of Data, school-level pre-kindergarten membership, 2018–19.

https://nces.ed.gov/ccd/Data/zip/ccd_sch_052_1819_1_1a_091019.zip

National Institute for Early Education Research

The State of Preschool 2019, for Head Start at ages 3 and 4.

<https://nieer.org/state-preschool-yearbook-2019>

The State(s) of Head Start and Early Head Start, 2023, for Early Head Start.

https://nieer.org/sites/default/files/2023-10/hs_fullreport.pdf

Child Care Aware of America

Kentucky licensed and certified provider registers, September 2018, republished as feature services.

https://services7.arcgis.com/s3vGpGobX9nz1LH3/arcgis/rest/services/Providers_Licensed/FeatureServer

Child Care Aware of Kentucky

2019 Kentucky Workforce Study, Table 20, for capacity and enrollment by age band.

<https://www.childcareawareky.org/research/>

New Jersey Department of Children and Families

Licensed child care centers, May 2019.

<https://data.nj.gov/resource/cru5-4rmm.csv>

New Jersey Department of Human Services

2021 Child Care Market Rate Survey, for the 67 percent center fill rate and the family child care provider universe.

<https://www.childcarenj.gov/Resources/Reports>

New York Office of Children and Family Services

Licensed child care providers, all 62 counties.

<https://data.ny.gov/resource/cb42-qumz.csv>

Provider counts and capacity by modality, 2015 to 2025, published separately for NYC and the rest of the state, for the vintage factors in sub-step 3.2. Held in the repository; the agency's site blocks automated retrieval.

New York City Department of Health and Mental Hygiene

Child care centers permitted under Article 47 of the city Health Code.

<https://data.cityofnewyork.us/resource/gy3q-4tzp.csv>

Virginia Department of Social Services

Licensed and approved child care providers, all approval routes.

<https://legacy.dss.virginia.gov/facility/search/cc2.cgi>

Virginia open data portal

Child Care Data Matrix, June 2020, published by VDSS, for the 0.90 vintage factor.

<https://data.virginia.gov/dataset/child-care-data-matrix-virginia-s-estimated-child-care-capacity-demand-june-2020>

Annenberg Institute, Brown University

Miller-Bains, Yu and Bassok, *Is Authorized Capacity a Good Measure of Child Care Providers' Current Capacity? New evidence from Virginia*, EdWorkingPaper 24-983, 2024.

<https://edworkingpapers.com/ai24-983>

This repository

`src/1_processing/1b_acs_processing.R`, where the filter `age < 5` appears in seven places, and `state_data/`, which holds the build and every raw source above.

5 Limits

1. The build cannot see unpaid relative care, which is the largest informal category nationally. That figure is a remainder rather than a measurement, and no administrative source would fix it. Only a restricted-access survey file with a state identifier would.
2. The mix carries the date of the register. The total is 2019 by construction, but how it divides across care types reflects whenever the state last published. Composition genuinely drifts, so a 2026 register shows too few family homes for 2019 even after the vintage correction in sub-step 3.2.

3. Unpaid Center-Based is an upper limit everywhere, because Head Start and state pre-K overlap by an unpublished amount and a program site may also hold a state licence.
4. Step 5 forces an exact match to the survey. That is a decision rather than a finding. The alternative is to let the registers stand and report the disagreement instead of removing it.
5. The simulator still expects seven rows, with center care split by price, and it rejects a file with six. Bridging the two is one mechanical operation that divides the center row on a national share. The build keeps that operation outside the method, so that it never appears to measure a quantity it cannot measure.